Chapter 8
Extended Variable Methods

In the previous chapter we described numerical methods for solving the equations of Langevin dynamics, a system of stochastic differential equations that sample the canonical ensemble. The methods allow us to compute averages of the form

$$\mathrm{Av}_\beta(\phi(\boldsymbol{q},\boldsymbol{p})) = \frac{\int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\rho_\beta(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega}{\int_{\mathcal{D}} \rho_\beta(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega},$$

where $\rho_\beta = \exp(-\beta H)$, and H is the system Hamiltonian. In this chapter we extend the range of systems that can be used to accomplish this task using methods based on the incorporation of auxiliary variables. We also consider schemes for sampling alternative thermodynamic ensembles, for example those defined by constant number of atoms N, pressure P and temperature T. In some of the methods, based on purely deterministic models, the ergodic property may be compromised, so an important issue is to design schemes that restore this. We will focus on methods that introduce random processes in direct contact with the auxiliary variables.

Besides canonical or configurational sampling, another common purpose of molecular simulation is the calculation of a dynamical (time-dependent) quantity such as the rearrangement of a molecular group over time or a diffusion rate. The only dynamical calculations that make sense in the context of molecular modelling, where trajectories are chaotic, are those that involve some sort of average. We consider this average in a thermodynamic sense and assume that the calculation can be formulated as the determination of the canonically weighted correlation between two quantities evaluated at different times, for example

$$C_\beta(\tau) = \frac{\int_{\mathcal{D}} \boldsymbol{A}(\boldsymbol{z}) \cdot \boldsymbol{B}(\mathscr{T}_\tau^H(\boldsymbol{z}))\rho_\beta \mathrm{d}\omega}{\int_{\mathcal{D}} \boldsymbol{A}(\boldsymbol{z}) \cdot \boldsymbol{B}(\boldsymbol{z})\rho_\beta \mathrm{d}\omega}.$$

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$\mathscr{F}_t^H$ represents the flow map of Hamiltonian dynamics. Thus $C(0) = 1$. At large τ, if the system is "mixing," and loses memory of its starting point, hence

$$\mathrm{Av}_\beta \left[\boldsymbol{A}(z) \cdot \boldsymbol{B}(\mathscr{F}_\tau^H(z))\right] \approx \mathrm{Av}_\beta \left[\boldsymbol{A}(z)\right] \cdot \mathrm{Av}_\beta \left[\boldsymbol{B}(z)\right]$$

is approximately independent of τ. Thus the correlation should approximately vanish at large time, provided the scalar product of the average of $\boldsymbol{A}$ and the average $\boldsymbol{B}$ vanishes. Computing an autocorrelation function $C_\beta(\tau)$ normally involves performing a stationary thermodynamic calculation in which the observable itself depends on a Hamiltonian dynamics calculation.

There is no reason for taking the averages solely with respect to ρ_β. We could in effect assume any prior distribution for the initial conditions. Our choice of Gibbs statistics in defining autocorrelation is made for purely practical reasons—it simplifies the study of the problem by allowing us to bring to bear the common techniques of thermodynamic sampling. In some cases, the Gibbs weight is ignored and the calculation is performed as an average two point correlation along a single trajectory. In some cases this calculation is performed using a single thermostatted molecular dynamics trajectory. Such a calculation is only justified when the thermostat is "weakly coupled" to the physical system. In this setup, the key to accurately computing averaged dynamics of molecular systems is to use thermostat techniques that minimally perturb the physical dynamics. We will briefly discuss how this may be achieved while maintaining the ergodic property.

8.1 Dynamics-Based Thermostat Techniques

Deterministic trajectories can sometimes be used to sample the canonical measure. A very simple technique of this type is Hamiltonian dynamics itself.

Let a Hamiltonian $H(\boldsymbol{q}, \boldsymbol{p})$ be given and let $\mathcal{L}$ be its Liouvillian

$$\mathcal{L} = \nabla_{\boldsymbol{p}} H \cdot \nabla_{\boldsymbol{q}} - \nabla_{\boldsymbol{q}} H \cdot \nabla_{\boldsymbol{p}}.$$

Recall from Chap. 5 that, as the dynamics automatically preserves functions of H, the Liouvillian satisfies $\mathcal{L}\varphi(H) = 0$. We also know that for Hamiltonian dynamics $\mathcal{L}$ is skew-adjoint, i.e., $\mathcal{L}^\dagger = -\mathcal{L}$ which implies that $\mathcal{L}^\dagger \psi(H) = 0$. Hence any density which is a function of H will be preserved, and this includes the canonical one $\psi(H) = \exp(-\beta H)$.

If we select a set of points at random from the canonical distribution, then initiate trajectories of Hamiltonian dynamics from each of these points, the points will remain Gibbs-distributed over time. If the paths themselves are ergodic on the surfaces of constant energy $H = \mathrm{E}$ then the collection of paths may provide a usable sampling of the canonical distribution. Such a sampling technique relies on having the means of choosing initial points from the canonical distribution; as this is the